

Bayes's theorem

From the definition (C.16) of conditional probability and the commutative law $A \cap B = B \cap A$, it follows that for two events A and B , each with nonzero probability,

$$\begin{aligned}\Pr\{A \cap B\} &= \Pr\{B\} \Pr\{A \mid B\} \\ &= \Pr\{A\} \Pr\{B \mid A\} .\end{aligned}\tag{C.18}$$

Solving for $\Pr\{A \mid B\}$, we obtain

$$\Pr\{A \mid B\} = \frac{\Pr\{A\} \Pr\{B \mid A\}}{\Pr\{B\}} ,\tag{C.19}$$

which is known as *Bayes's theorem*. The denominator $\Pr\{B\}$ is a normalizing constant, which we can reformulate as follows. Since $B = (B \cap A) \cup (B \cap \bar{A})$, and since $B \cap A$ and $B \cap \bar{A}$ are mutually exclusive events,

$$\begin{aligned}\Pr\{B\} &= \Pr\{B \cap A\} + \Pr\{B \cap \bar{A}\} \\ &= \Pr\{A\} \Pr\{B \mid A\} + \Pr\{\bar{A}\} \Pr\{B \mid \bar{A}\} .\end{aligned}$$

Substituting into equation (C.19) produces an equivalent form of Bayes's theorem:

$$\Pr\{A \mid B\} = \frac{\Pr\{A\} \Pr\{B \mid A\}}{\Pr\{A\} \Pr\{B \mid A\} + \Pr\{\bar{A}\} \Pr\{B \mid \bar{A}\}} .\tag{C.20}$$

Bayes's theorem can simplify the computing of conditional probabilities. For example, suppose that you have a fair coin and a biased coin that always comes up heads. Run an experiment consisting of three independent events: choose one of the two coins at random, flip that coin once, and then flip it again. Suppose that the coin you have chosen comes up heads both times. What is the probability that it's the biased coin?

Bayes's theorem solves this problem. Let A be the event that you choose the biased coin, and let B be the event that the chosen coin comes up heads both times. We wish to determine $\Pr\{A \mid B\}$, knowing that $\Pr\{A\} = 1/2$, $\Pr\{B \mid A\} = 1$, $\Pr\{\bar{A}\} = 1/2$, and $\Pr\{B \mid \bar{A}\} = 1/4$. Thus we have

$$\begin{aligned}\Pr\{A \mid B\} &= \frac{(1/2) \cdot 1}{(1/2) \cdot 1 + (1/2) \cdot (1/4)} \\ &= 4/5 .\end{aligned}$$

Exercises**C.2-1**

Professor Rosencrantz flips a fair coin twice. Professor Guildenstern flips a fair coin once. What is the probability that Professor Rosencrantz obtains strictly more heads than Professor Guildenstern?

C.2-2

Prove *Boole's inequality*: For any finite or countably infinite sequence of events $A_1, A_2, \dots$,

$$\Pr\{A_1 \cup A_2 \cup \dots\} \leq \Pr\{A_1\} + \Pr\{A_2\} + \dots . \quad (\text{C.21})$$

C.2-3

You shuffle a deck of 10 cards, each bearing a distinct number from 1 to 10, in order to mix the cards thoroughly. You then remove three cards, one at a time, from the deck. What is the probability that the three cards you select are in sorted (increasing) order?

C.2-4

Prove that

$$\Pr\{A \mid B\} + \Pr\{\overline{A} \mid B\} = 1 .$$

C.2-5

Prove that for any collection of events $A_1, A_2, \dots, A_n$,

$$\Pr\{A_1 \cap A_2 \cap \dots \cap A_n\} = \Pr\{A_1\} \cdot \Pr\{A_2 \mid A_1\} \cdot \Pr\{A_3 \mid A_1 \cap A_2\} \cdots \Pr\{A_n \mid A_1 \cap A_2 \cap \dots \cap A_{n-1}\} . \quad (\text{C.22})$$

★ C.2-6

Show how to construct a set of n events that are pairwise independent but such that no subset of $k > 2$ of them is mutually independent.

★ C.2-7

Two events A and B are *conditionally independent*, given C , if

$$\Pr\{A \cap B \mid C\} = \Pr\{A \mid C\} \cdot \Pr\{B \mid C\} .$$

Give a simple but nontrivial example of two events that are not independent but are conditionally independent given a third event.

★ C.2-8

Professor Gore teaches a music class on rhythm in which three students—Jeff, Tim, and Carmine—are in danger of failing. Professor Gore tells the three that one of

them will pass the course and the other two will fail. Carmine asks Professor Gore privately which of Jeff and Tim will fail, arguing that since he already knows at least one of them will fail, the professor won't be revealing any information about Carmine's outcome. In a breach of privacy law, Professor Gore tells Carmine that Jeff will fail. Carmine feels somewhat relieved now, figuring that either he or Tim will pass, so that his probability of passing is now $1/2$. Is Carmine correct, or is his chance of passing still $1/3$? Explain.

C.3 Discrete random variables

A **(discrete) random variable** X is a function from a finite or countably infinite sample space S to the real numbers. It associates a real number with each possible outcome of an experiment, which allows us to work with the probability distribution induced on the resulting set of numbers. Random variables can also be defined for uncountably infinite sample spaces, but they raise technical issues that are unnecessary to address for our purposes. Therefore we'll assume that random variables are discrete.

For a random variable X and a real number x , we define the event $X = x$ to be $\{s \in S : X(s) = x\}$, and thus

$$\Pr\{X = x\} = \sum_{s \in S: X(s)=x} \Pr\{s\}.$$

The function

$$f(x) = \Pr\{X = x\}$$

is the **probability density function** of the random variable X . From the probability axioms, $\Pr\{X = x\} \geq 0$ and $\sum_x \Pr\{X = x\} = 1$.

As an example, consider the experiment of rolling a pair of ordinary, 6-sided dice. There are 36 possible outcomes in the sample space. Assume that the probability distribution is uniform, so that each outcome $s \in S$ is equally likely: $\Pr\{s\} = 1/36$. Define the random variable X to be the *maximum* of the two values showing on the dice. We have $\Pr\{X = 3\} = 5/36$, since X assigns a value of 3 to 5 of the 36 possible outcomes, namely, (1, 3), (2, 3), (3, 3), (3, 2), and (3, 1).

We can define several random variables on the same sample space. If X and Y are random variables, the function

$$f(x, y) = \Pr\{X = x \text{ and } Y = y\}$$

is the **joint probability density function** of X and Y . For a fixed value y ,

$$\Pr\{Y = y\} = \sum_x \Pr\{X = x \text{ and } Y = y\} ,$$

and similarly, for a fixed value x ,

$$\Pr\{X = x\} = \sum_y \Pr\{X = x \text{ and } Y = y\} .$$

Using the definition (C.16) of conditional probability on page 1187, we have

$$\Pr\{X = x \mid Y = y\} = \frac{\Pr\{X = x \text{ and } Y = y\}}{\Pr\{Y = y\}} .$$

We define two random variables X and Y to be *independent* if for all x and y , the events $X = x$ and $Y = y$ are independent or, equivalently, if for all x and y , we have $\Pr\{X = x \text{ and } Y = y\} = \Pr\{X = x\} \Pr\{Y = y\}$.

Given a set of random variables defined over the same sample space, we can define new random variables as sums, products, or other functions of the original variables.

Expected value of a random variable

The simplest, and often the most useful, summary of the distribution of a random variable is the “average” of the values it takes on. The *expected value* (or, synonymously, *expectation* or *mean*) of a discrete random variable X is

$$E[X] = \sum_x x \cdot \Pr\{X = x\} , \tag{C.23}$$

which is well defined if the sum is finite or converges absolutely. Sometimes the expectation of X is denoted by μ_X or, when the random variable is apparent from context, simply by μ .

Consider a game in which you flip two fair coins. You earn \$3 for each head but lose \$2 for each tail. The expected value of the random variable X representing your earnings is

$$\begin{aligned} E[X] &= 6 \cdot \Pr\{2 \text{ H's}\} + 1 \cdot \Pr\{1 \text{ H}, 1 \text{ T}\} - 4 \cdot \Pr\{2 \text{ T's}\} \\ &= 6 \cdot (1/4) + 1 \cdot (1/2) - 4 \cdot (1/4) \\ &= 1 . \end{aligned}$$

Linearity of expectation says that the expectation of the sum of two random variables is the sum of their expectations, that is,

$$E[X + Y] = E[X] + E[Y] , \tag{C.24}$$

whenever $E[X]$ and $E[Y]$ are defined. Linearity of expectation applies to a broad range of situations, holding even when X and Y are not independent. It also extends to finite and absolutely convergent summations of expectations. Linearity of expectation is the key property that enables us to perform probabilistic analyses by using indicator random variables (see Section 5.2).

If X is any random variable, any function $g(x)$ defines a new random variable $g(X)$. If the expectation of $g(X)$ is defined, then

$$E[g(X)] = \sum_x g(x) \cdot \Pr\{X = x\} .$$

Letting $g(x) = ax$, we have for any constant a ,

$$E[aX] = aE[X] . \quad (\text{C.25})$$

Consequently, expectations are linear: for any two random variables X and Y and any constant a ,

$$E[aX + Y] = aE[X] + E[Y] . \quad (\text{C.26})$$

When two random variables X and Y are independent and each has a defined expectation,

$$\begin{aligned} E[XY] &= \sum_x \sum_y xy \cdot \Pr\{X = x \text{ and } Y = y\} \\ &= \sum_x \sum_y xy \cdot \Pr\{X = x\} \Pr\{Y = y\} \quad (\text{by independence of } X \text{ and } Y) \\ &= \left(\sum_x x \cdot \Pr\{X = x\} \right) \left(\sum_y y \cdot \Pr\{Y = y\} \right) \\ &= E[X]E[Y] \quad (\text{by equation (C.23)}) . \end{aligned}$$

In general, when n random variables $X_1, X_2, \dots, X_n$ are mutually independent,

$$E[X_1 X_2 \cdots X_n] = E[X_1] E[X_2] \cdots E[X_n] . \quad (\text{C.27})$$

When a random variable X takes on values from the set of natural numbers $\mathbb{N} = \{0, 1, 2, \dots\}$, we have a nice formula for its expectation:

$$\begin{aligned} E[X] &= \sum_{i=0}^{\infty} i \cdot \Pr\{X = i\} \\ &= \sum_{i=0}^{\infty} i \cdot (\Pr\{X \geq i\} - \Pr\{X \geq i + 1\}) \\ &= \sum_{i=1}^{\infty} \Pr\{X \geq i\} , \end{aligned} \quad (\text{C.28})$$

since each term $\Pr\{X \geq i\}$ is added in i times and subtracted out $i - 1$ times (except $\Pr\{X \geq 0\}$, which is added in 0 times and not subtracted out at all).

A function $f(x)$ is **convex** if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \quad (\text{C.29})$$

for all x and y and for all $0 \leq \lambda \leq 1$. **Jensen's inequality** says that when a convex function $f(x)$ is applied to a random variable X ,

$$E[f(X)] \geq f(E[X]), \quad (\text{C.30})$$

provided that the expectations exist and are finite.

Variance and standard deviation

The expected value of a random variable does not express how “spread out” the variable’s values are. For example, consider random variables X and Y for which $\Pr\{X = 1/4\} = \Pr\{X = 3/4\} = 1/2$ and $\Pr\{Y = 0\} = \Pr\{Y = 1\} = 1/2$. Then both $E[X]$ and $E[Y]$ are $1/2$, yet the actual values taken on by Y are further from the mean than the actual values taken on by X .

The notion of variance mathematically expresses how far from the mean a random variable’s values are likely to be. The **variance** of a random variable X with mean $E[X]$ is

$$\begin{aligned} \text{Var}[X] &= E[(X - E[X])^2] \\ &= E[X^2 - 2XE[X] + E^2[X]] \\ &= E[X^2] - 2E[XE[X]] + E^2[X] \\ &= E[X^2] - 2E^2[X] + E^2[X] \\ &= E[X^2] - E^2[X]. \end{aligned} \quad (\text{C.31})$$

To justify the equation $E[E^2[X]] = E^2[X]$, note that because $E[X]$ is a real number and not a random variable, so is $E^2[X]$. The equation $E[XE[X]] = E^2[X]$ follows from equation (C.25), with $a = E[X]$. Rewriting equation (C.31) yields an expression for the expectation of the square of a random variable:

$$E[X^2] = \text{Var}[X] + E^2[X]. \quad (\text{C.32})$$

The variance of a random variable X and the variance of aX are related (see Exercise C.3-10):

$$\text{Var}[aX] = a^2 \text{Var}[X].$$

When X and Y are independent random variables,

$$\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y].$$

In general, if n random variables $X_1, X_2, \dots, X_n$ are pairwise independent, then

$$\text{Var} \left[\sum_{i=1}^n X_i \right] = \sum_{i=1}^n \text{Var} [X_i] . \quad (\text{C.33})$$

The **standard deviation** of a random variable X is the nonnegative square root of the variance of X . The standard deviation of a random variable X is sometimes denoted σ_X or simply σ when the random variable X is understood from context. With this notation, the variance of X is denoted σ^2 .

Exercises

C.3-1

You roll two ordinary, 6-sided dice. What is the expectation of the sum of the two values showing? What is the expectation of the maximum of the two values showing?

C.3-2

An array $A[1 : n]$ contains n distinct numbers that are randomly ordered, with each permutation of the n numbers being equally likely. What is the expectation of the index of the maximum element in the array? What is the expectation of the index of the minimum element in the array?

C.3-3

A carnival game consists of three dice in a cage. A player can bet a dollar on any of the numbers 1 through 6. The cage is shaken, and the payoff is as follows. If the player's number doesn't appear on any of the dice, the player loses the dollar. Otherwise, if the player's number appears on exactly k of the three dice, for $k = 1, 2, 3$, the player keeps the dollar and wins k more dollars. What is the expected gain from playing the carnival game once?

C.3-4

Argue that if X and Y are nonnegative random variables, then

$$\text{E} [\max \{X, Y\}] \leq \text{E} [X] + \text{E} [Y] .$$

★ C.3-5

Let X and Y be independent random variables. Prove that $f(X)$ and $g(Y)$ are independent for any choice of functions f and g .

★ C.3-6

Let X be a nonnegative random variable, and suppose that $E[X]$ is well defined. Prove **Markov's inequality**:

$$\Pr\{X \geq t\} \leq E[X] / t \quad (\text{C.34})$$

for all $t > 0$.

★ C.3-7

Let S be a sample space, and let X and X' be random variables such that $X(s) \geq X'(s)$ for all $s \in S$. Prove that for any real constant t ,

$$\Pr\{X \geq t\} \geq \Pr\{X' \geq t\}.$$

C.3-8

Which is larger: the expectation of the square of a random variable, or the square of its expectation?

C.3-9

Show that for any random variable X that takes on only the values 0 and 1, we have $\text{Var}[X] = E[X]E[1 - X]$.

C.3-10

Prove that $\text{Var}[aX] = a^2\text{Var}[X]$ from the definition (C.31) of variance.

C.4 The geometric and binomial distributions

A **Bernoulli trial** is an experiment with only two possible outcomes: **success**, which occurs with probability p , and **failure**, which occurs with probability $q = 1 - p$. A coin flip serves as an example where, depending on your point of view, heads equates to success and tails to failure. When we speak of **Bernoulli trials** collectively, we mean that the trials are mutually independent and, unless we specifically say otherwise, that each has the same probability p for success. Two important distributions arise from Bernoulli trials: the geometric distribution and the binomial distribution.

The geometric distribution

Consider a sequence of Bernoulli trials, each with a probability p of success and a probability $q = 1 - p$ of failure. How many trials occur before a success? Define the random variable X to be the number of trials needed to obtain a success. Then X has values in the range $\{1, 2, \dots\}$, and for $k \geq 1$,

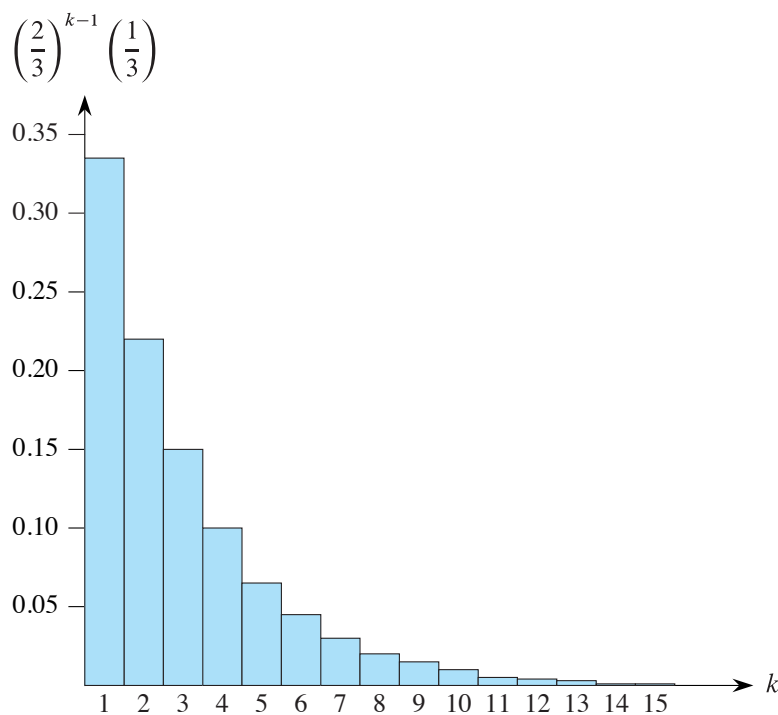


Figure C.1 A geometric distribution with probability $p = 1/3$ of success and a probability $q = 1 - p$ of failure. The expectation of the distribution is $1/p = 3$.

$$\Pr\{X = k\} = q^{k-1} p, \quad (\text{C.35})$$

since $k - 1$ failures occur before the first success. A probability distribution satisfying equation (C.35) is said to be a *geometric distribution*. Figure C.1 illustrates such a distribution.

Assuming that $q < 1$, we can calculate the expectation of a geometric distribution:

$$\begin{aligned}
 E[X] &= \sum_{k=1}^{\infty} k q^{k-1} p \\
 &= \frac{p}{q} \sum_{k=0}^{\infty} k q^k \\
 &= \frac{p}{q} \cdot \frac{q}{(1-q)^2} \quad (\text{by equation (A.11) on page 1142}) \\
 &= \frac{p}{q} \cdot \frac{q}{p^2} \\
 &= 1/p.
 \end{aligned} \quad (\text{C.36})$$

Thus, on average, it takes $1/p$ trials before a success occurs, an intuitive result. As Exercise C.4-3 asks you to show, the variance is

$$\text{Var}[X] = q/p^2. \quad (\text{C.37})$$

As an example, suppose that you repeatedly roll two dice until you obtain either a seven or an eleven. Of the 36 possible outcomes, 6 yield a seven and 2 yield an eleven. Thus, the probability of success is $p = 8/36 = 2/9$, and you'd have to roll $1/p = 9/2 = 4.5$ times on average to obtain a seven or eleven.

The binomial distribution

How many successes occur during n Bernoulli trials, where a success occurs with probability p and a failure with probability $q = 1 - p$? Define the random variable X to be the number of successes in n trials. Then X has values in the range $\{0, 1, \dots, n\}$, and for $k = 0, 1, \dots, n$,

$$\Pr\{X = k\} = \binom{n}{k} p^k q^{n-k}, \quad (\text{C.38})$$

since there are $\binom{n}{k}$ ways to pick which k of the n trials are successes, and the probability that each occurs is $p^k q^{n-k}$. A probability distribution satisfying equation (C.38) is said to be a **binomial distribution**. For convenience, we define the family of binomial distributions using the notation

$$b(k; n, p) = \binom{n}{k} p^k (1 - p)^{n-k}. \quad (\text{C.39})$$

Figure C.2 illustrates a binomial distribution. The name “binomial” comes from the right-hand side of equation (C.38) being the k th term of the expansion of $(p + q)^n$. Consequently, since $p + q = 1$, equation (C.4) on page 1181 gives

$$\sum_{k=0}^n b(k; n, p) = 1, \quad (\text{C.40})$$

as axiom 2 of the probability axioms requires.

We can compute the expectation of a random variable having a binomial distribution from equations (C.9) and (C.40). Let X be a random variable that follows the binomial distribution $b(k; n, p)$, and let $q = 1 - p$. The definition of expectation gives